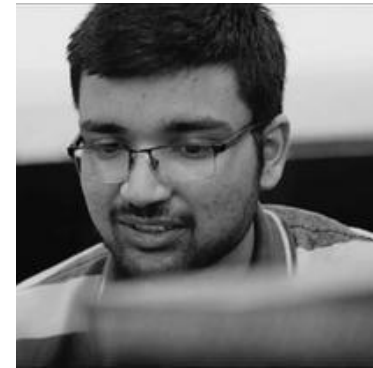


# TechForChange

## COHORT 4





Adarsh Goyal  
**Founder & CEO**



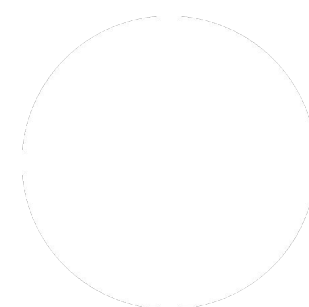
Sneha Gupta  
**Co-Founder & COO**



Naman Dadhich  
**Web Developer**



Sarthak Kakar  
**Web Designer**





# The Problem: STEM Is Still Visual-First

01.



**Graphs, geometry, circuits, and equations are hard to understand through Braille alone.**

02.



**Current tactile tools and Braille books are bulky, heavy, and difficult to carry, making STEM learning less portable and less convenient.**

03.



**Audio can describe a shape, but it cannot help students physically feel graphs or diagrams.**

04.



**Existing tactile displays and specialized learning tools are unaffordable for most students and schools.**



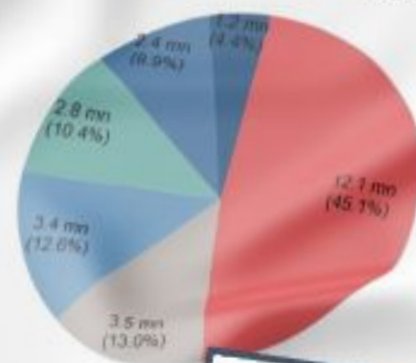


Braille literacy is low around the globe. In the United Kingdom, a more developed nation than India, the literacy rate for blind people is only 4%. The low number correlates with low employment rates for blind people. Existing methods of teaching are part of the reason the number is so low; it takes one student one

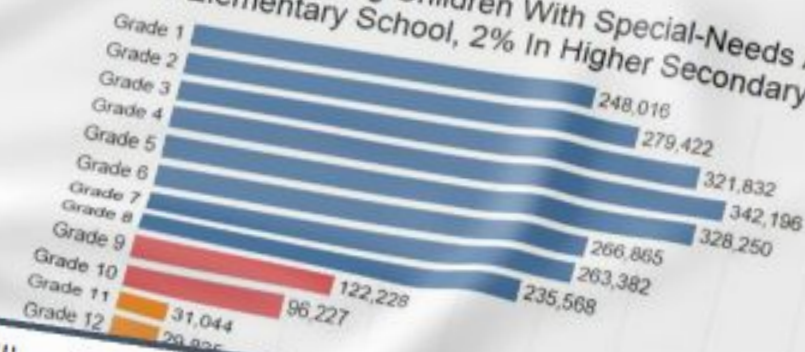
### One education policy does not fit all

As things stand, 45% of India's disabled population is illiterate, according to [Census 2011](#), compared to 26% of all Indians. Of persons with disabilities who are educated, 59% complete Class X, compared to 67% of the general population.

45% Of Indians With Special Needs Are Illiterate



89% School-Going Children With Special-Needs Are In Elementary School, 2% In Higher Secondary



Of the world's 37 million blind people, an estimated 90% live in developing countries. About 15 million of them live in India, a number that has doubled since 2007 to make it the highest number in any country. However, the braille literacy rate there is only 1%, far lower than the regular literacy rate of 77.7%. This

### BORGEN Magazine

THE GOOD NEWS

BUSINESS & NEW MARKETS

POLITICS

CELEB

TECH

WORLD

HUMANITY, POLITICS & YOU

## FIGHTING THE DENIAL OF EDUCATION FOR BLIND PEOPLE IN INDIA



Vasudha M, a tailor and mother of a blind daughter exclaimed: Most special educators and general teachers in these integrated schools themselves aren't adept in Braille. She struggles to read the texts. Although there are assistive devices, they aren't accessible or available to everyone. In Tamil Nadu, government schools provide study materials in Braille only up to class 8.

If inadequacy is an issue, the increasing price of Braille books also burdens the community. The price of the books is over 30-40 per cent higher than regular editions due to the high cost of printing.

Muniyappan, a History professor and member of the College Students and Graduates Association for the Blind (CSGAB), explains how a lack of Braille books can determine a visually impaired person's future.

When we lack Braille books for school-level subjects, imagine wanting to study Engineering but having to learn the voluminous syllabus through audio lessons. That is why most students take up Arts and Humanities and do not branch out to other professional streams. The failure to provide access creates barriers for education and employment, and even undermines an individual's capability," he elucidates.

## Six dots and social inclusion

She opens her eyes and looks at her daughter, who is now intently focusing on her mother's voice.

At least two visually-challenged students have been in the news recently for approaching the courts – Kritika Purohit filed a case in the Bombay High Court to be allowed to study physiotherapy and Reshma Dileep approached the Kerala High Court to be allowed to study science beyond secondary school.



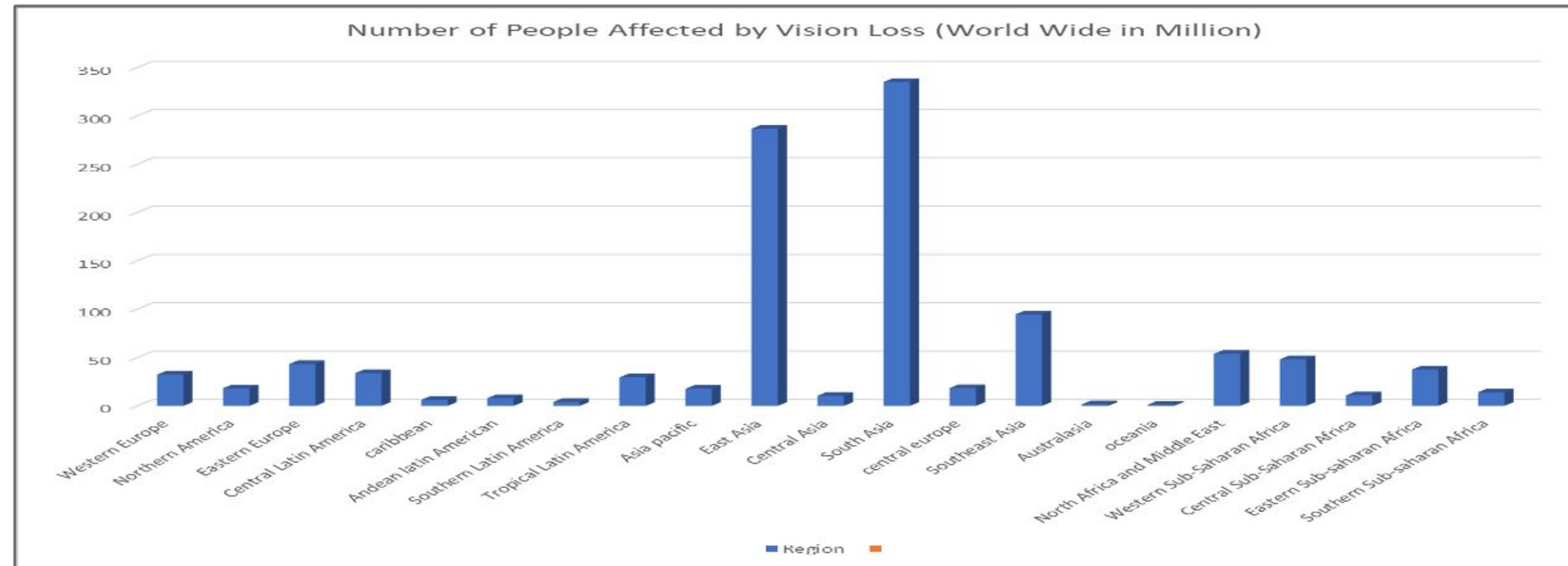
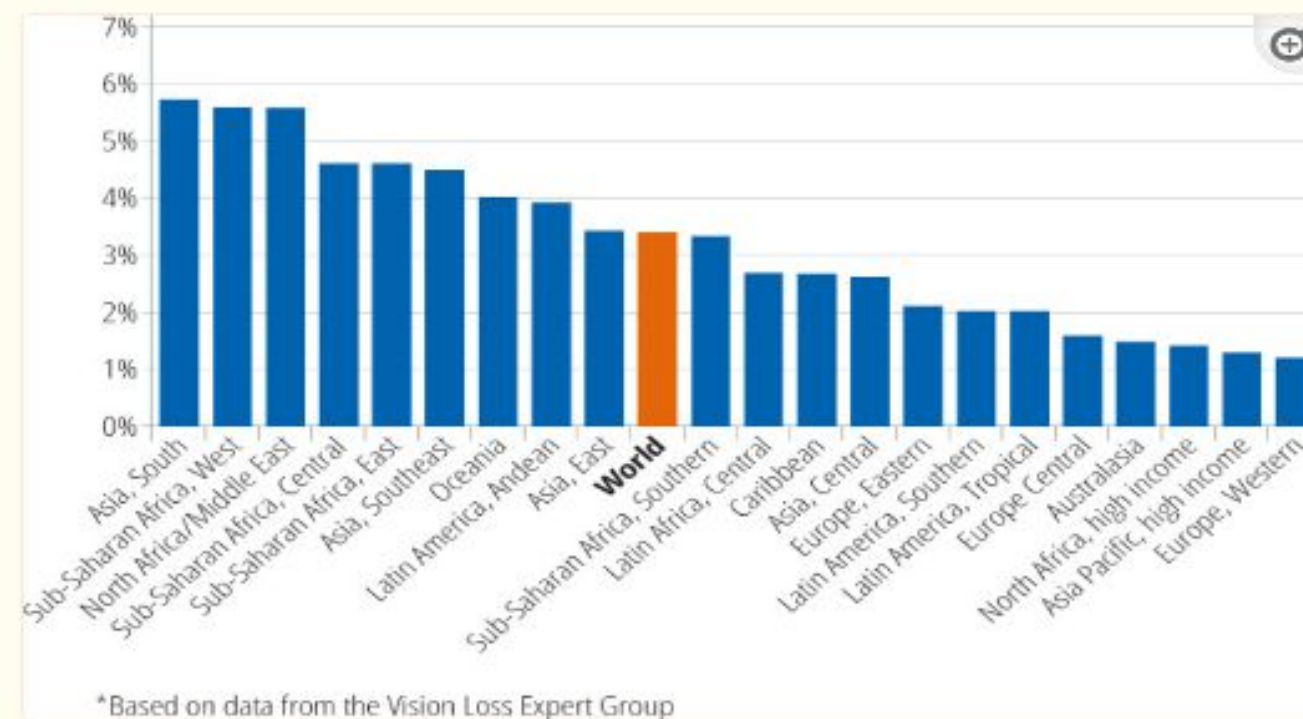


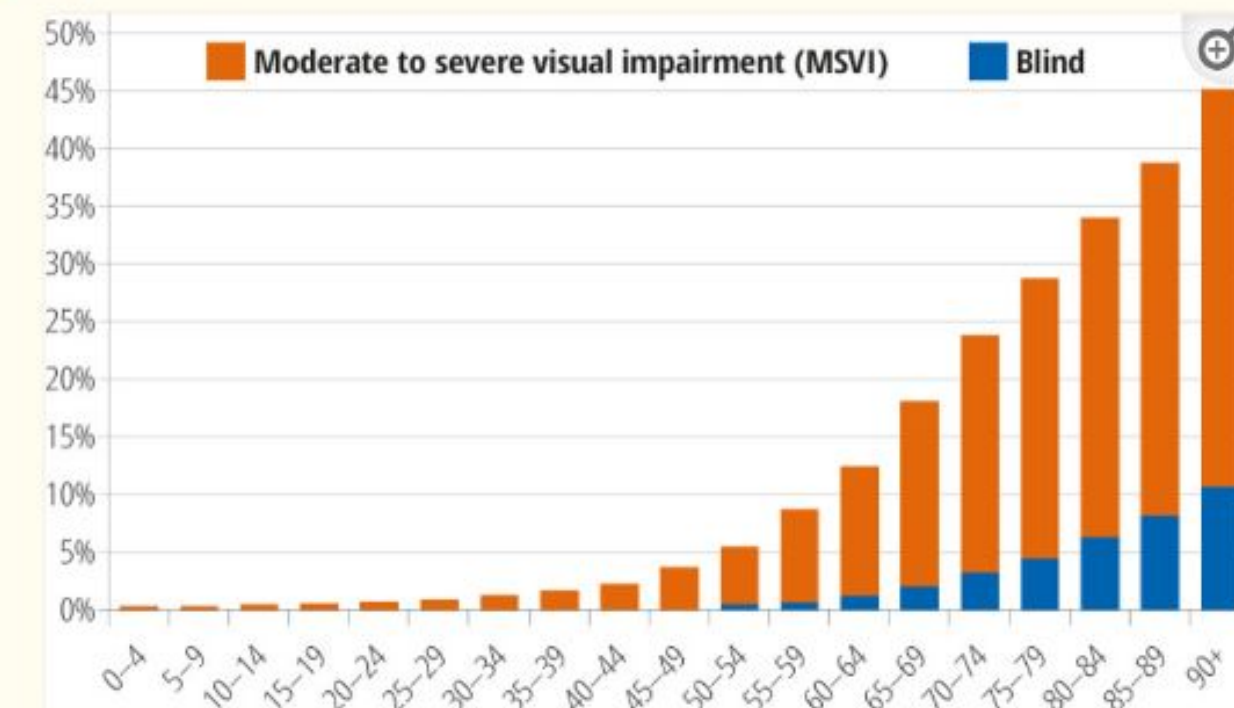
Figure 3



Age-standardised prevalence of visual impairment across the 21 regions\*

\*Based on data from the Vision Loss Expert Group

Figure 2

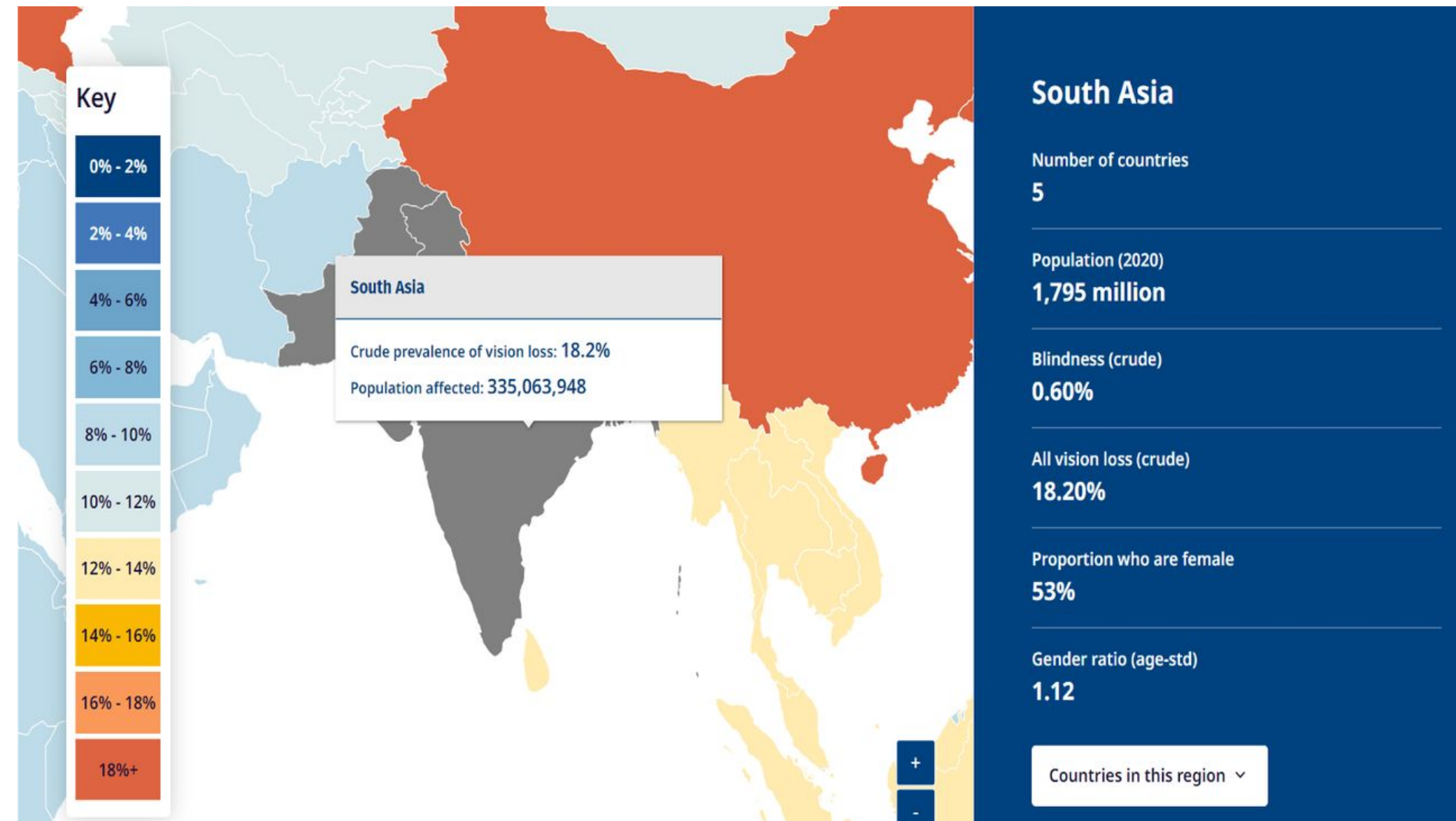


The global prevalence of blindness and moderate to severe visual impairment (MSVI) in women\*



**With 1.18M+ school-age visually impaired learners in India and 67M+ globally, Viisar addresses a large, underserved market with an affordable tactile STEM solution.**

| Age                                                     | India         | Worldwide     |
|---------------------------------------------------------|---------------|---------------|
| Preschool age<br>(0–6 years)                            | ~0.42 million | ~31.8 million |
| School-age<br>(5–19 years)                              | ~1.18 million | ~67.4 million |
| India – College-age /<br>young adults (20–24<br>years): | ~0.39 million | ~20.6million  |







# User interviews/survey insights

[User-Validation \(1080p, h264\).mp4](#)







## Target Users:

We visited **Shri Sai Nethraheen Sanstha** and **Blind Person Association** in Uttam Nagar and we tested our product on visually impaired individuals.

## Feedbacks Received Based on Interviews and Testing Sessions:

25+ Users Tested

Right now we mostly rely on audiobooks, but they don't help us feel shapes. Geometry books and physical models are also heavy and hard to carry. Your device solves this problem because we can feel different shapes on a single device, and it is much easier to carry.

Rajesh

This product helps us imagine shapes in our mind. For ex- it helps us understand how A looks, how a triangle feels, or what other figures are like. For blind students, it is very helpful because it allows us to connect more with what we are studying. With this device, understanding shapes and figures becomes much easier.

Krishna

The device feels a bit large right now. If the size could be reduced, it would be better because then we could feel the whole image using just our palm. That would make it more comfortable and easier to use.

Rajesh





[New Hardware - VIISAR\(1080p, h264\).mp4](#)



# Current Solutions vs Existing Gaps

| Factor                  | Graphiti             | Dot Pad                                   | BrailleMe                      |
|-------------------------|----------------------|-------------------------------------------|--------------------------------|
| Cost accessibility      | Low ( ~\$20,000 USD) | Low ( ~\$10,000 USD)                      | High (~ \$1,000 USD)           |
| Multi-line support      | Yes                  | Yes                                       | No, mainly 20-cell single-line |
| STEM suitability        | Strong but expensive | Strong but expensive                      | Limited                        |
| Tactile graphics        | Strong               | Story                                     | Weak / not primary             |
| Portability             | Medium               | Medium; Dot Pad X is more tablet-like     | High                           |
| Local manufacturing     | Low                  | Low                                       | Medium, India-origin product   |
| Repairability           | Low to medium        | Low to medium                             | Medium                         |
| Open-source scalability | Limited              | Limited                                   | Limited                        |
| Best suited for         | Premium institutions | Premium accessibility labs / institutions | Individual Braille readers     |





# Introducing Viisar:

## Affordable tactile STEM learning:

A lightweight, modular refreshable tactile display that converts graphs, geometry, and complex mathematical concepts into touch-based learning.

## Built for accessibility and portability:

It directly solves the affordability, multi-line display, and portability gaps—making Braille and STEM content accessible to millions of visually impaired learners and professionals at a fraction of the cost.

## Scalable and cost-effective:

Reduces the cost of tactile display technology from **\$10,000–\$20,000 to \$500–\$1,500**, while enabling local manufacturing, repair, and open-source scalability.







# SOFTWARE SERVICES

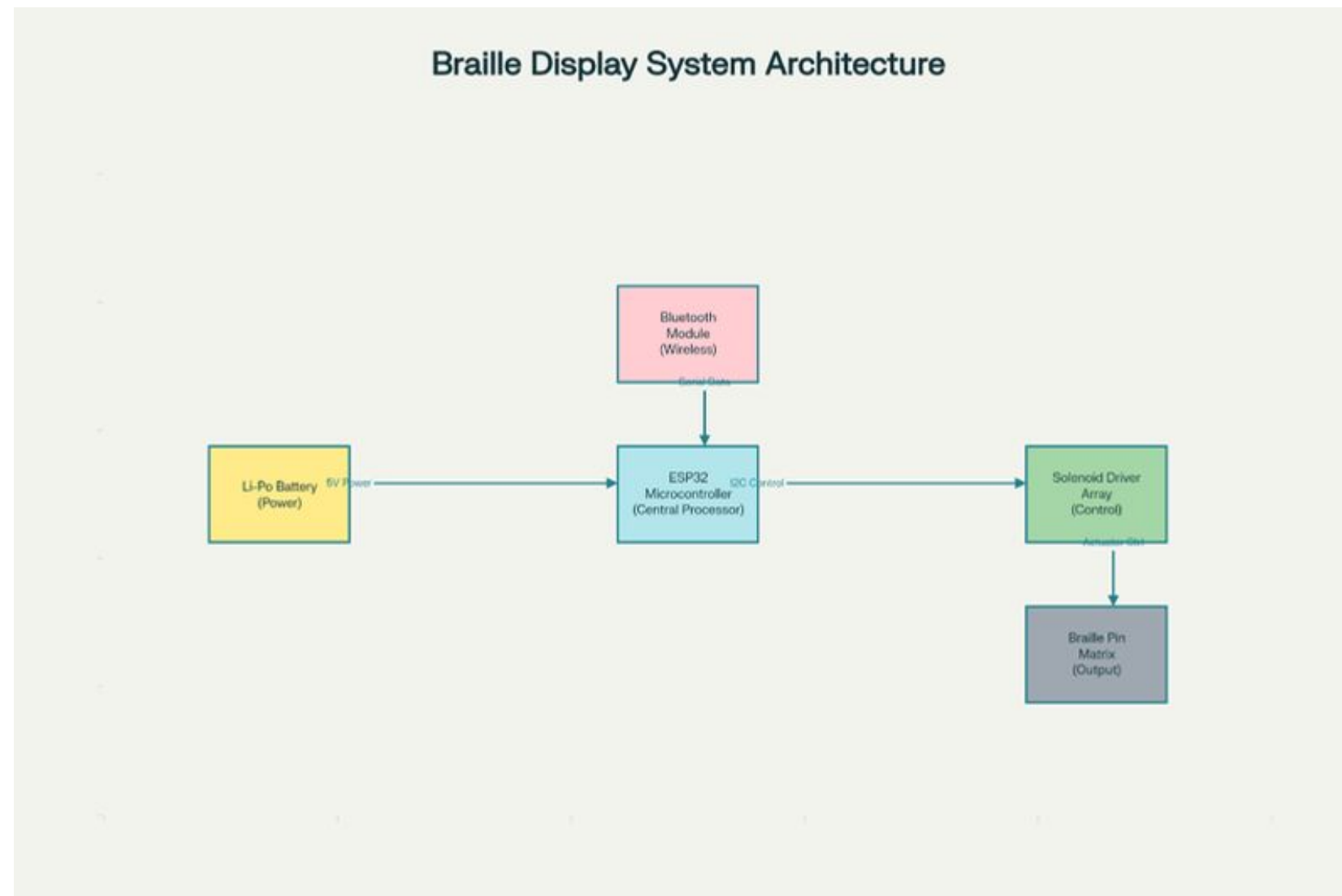
- Interactive Learning Modules
- AI-Powered Feedback System
- Portable Tactile-Compatible Interface
- Multilingual Support

Website Link: <https://viisar.vercel.app/>



**We help visually impaired students achieve confidence and inclusion in STEM through affordable, touch-based learning tools.**

## System Architecture

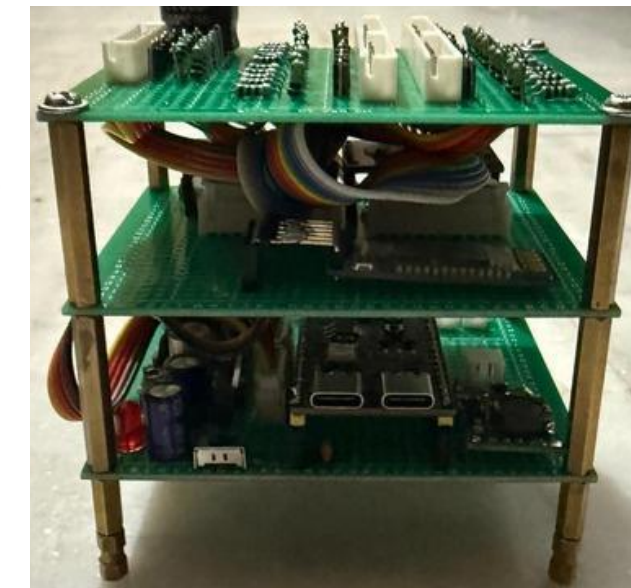


Bluetooth-connected app drives a tactile pin-matrix via ESP32 + solenoid drivers.

## Hardware Prototype



Final Prototype



Internal Circuitry

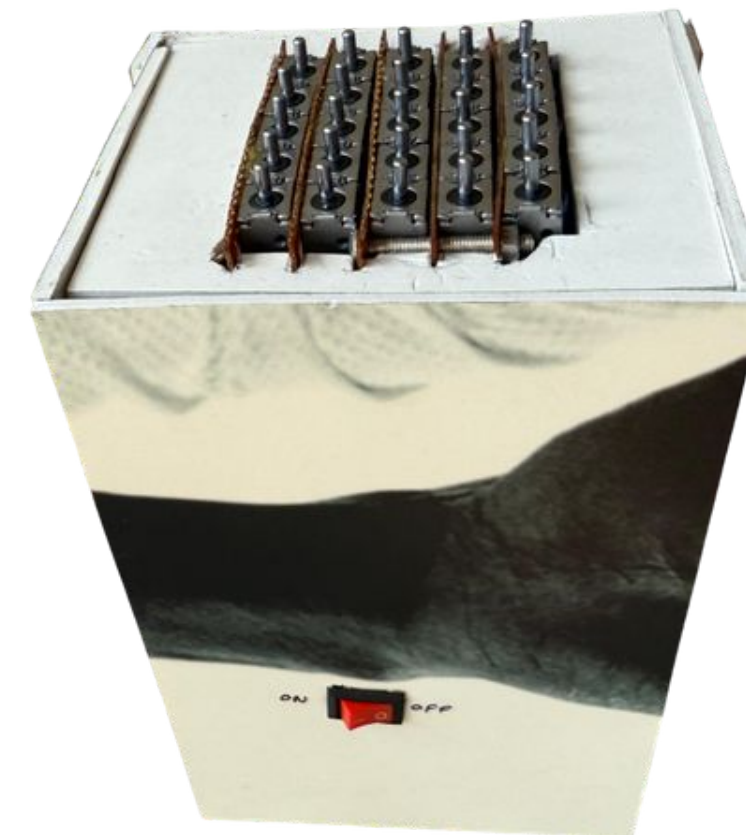
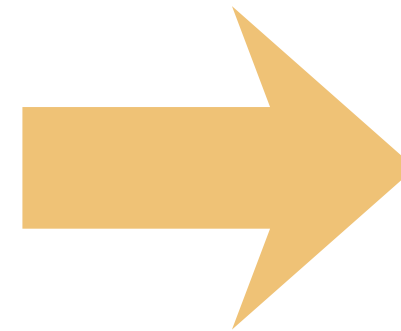




Based on the feedback we received, we redesigned the hardware and made it smaller in size. This allows users to easily feel the images using their palm, and the device can now comfortably fit in the hand.



Before

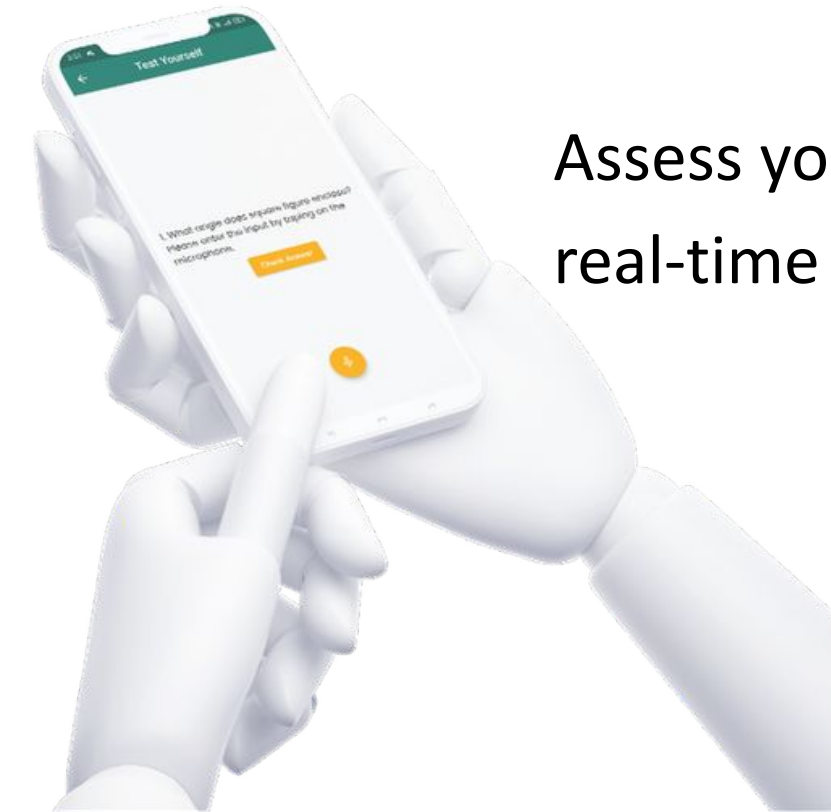


After





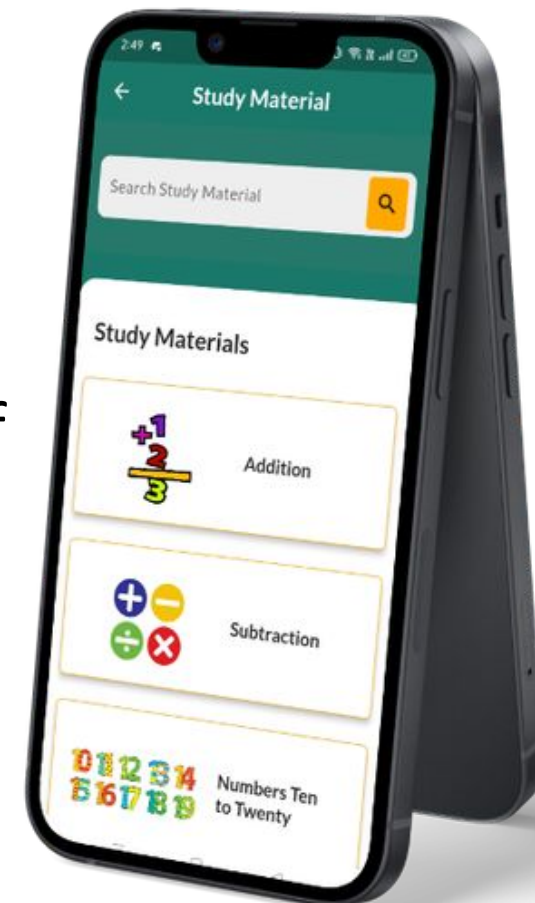
Use a Bluetooth-connected app and hardware to visually trace and comprehend figure geometry in real time



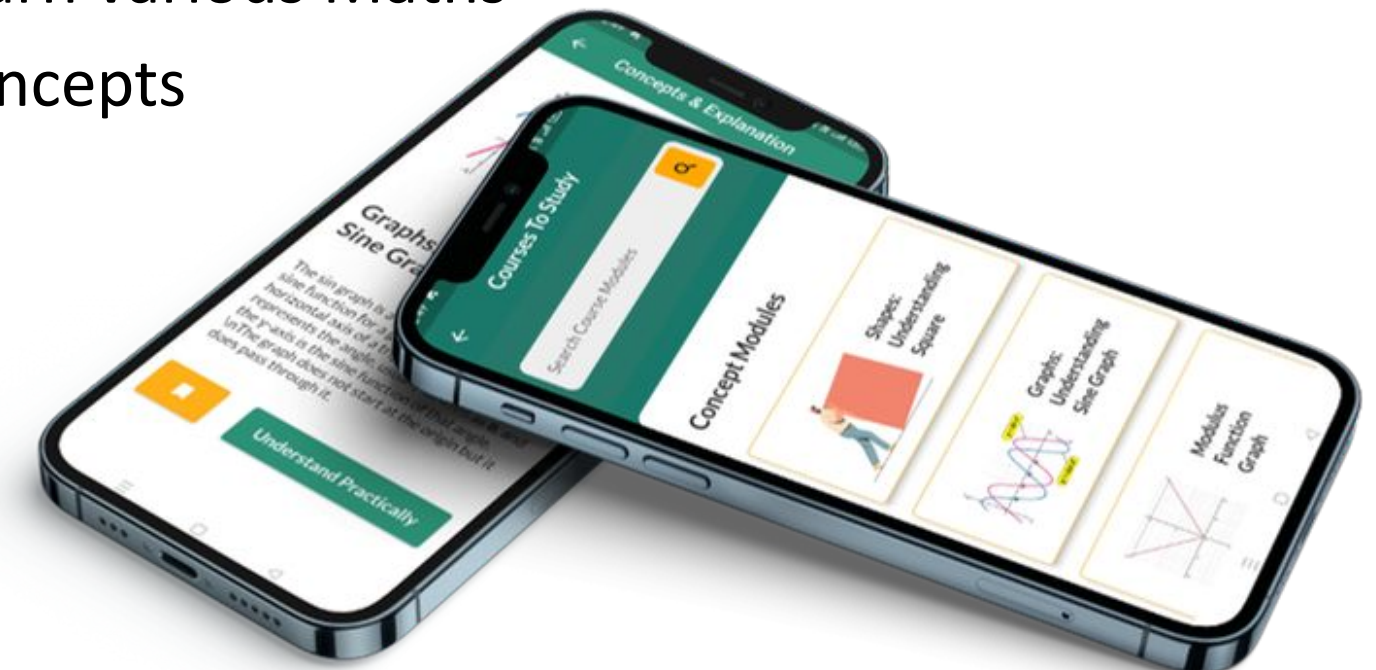
Assess your knowledge using real-time audio responses.

## Key Features

Mobile app with voice read-aloud feature offering a diverse range of STEM study materials.



Explore and Learn various Maths and Science Concepts







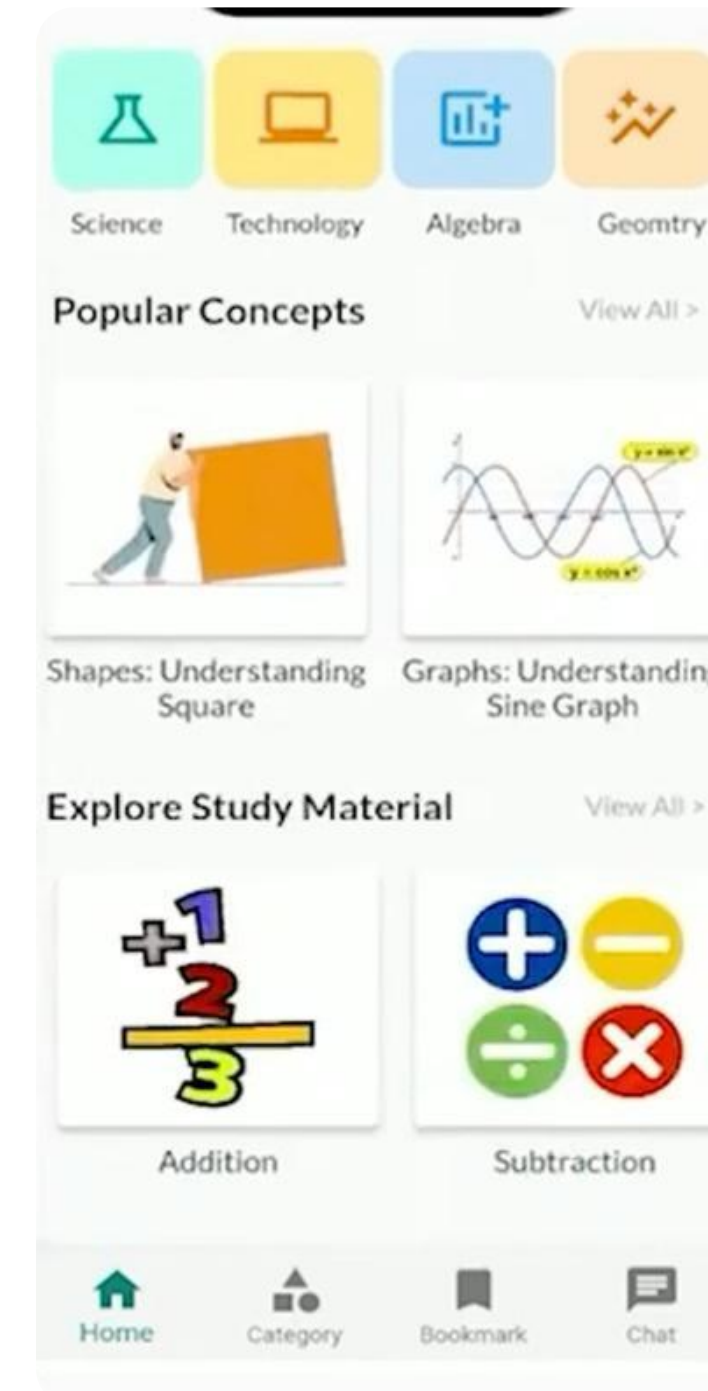
## Screens



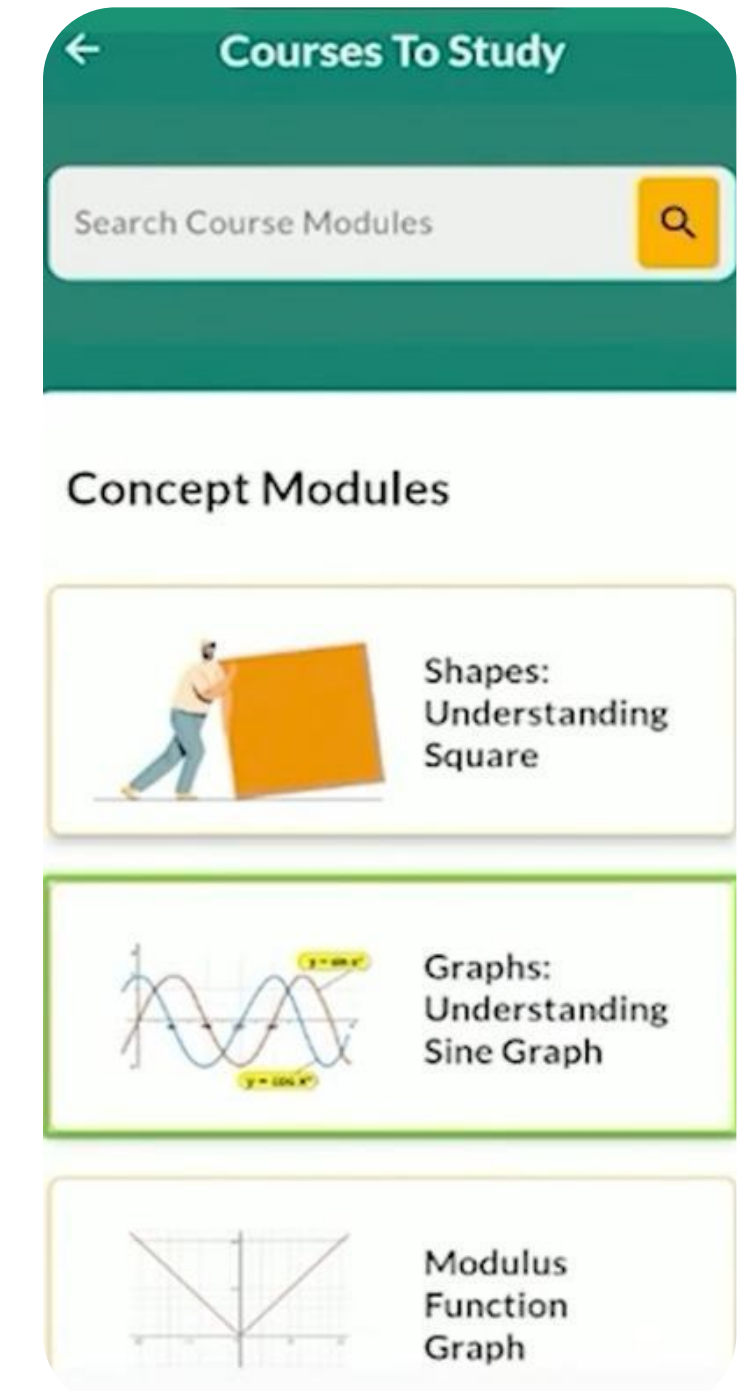
Starting with the app



Available options to learn, practice and test



Offering a diverse range of STEM study materials.

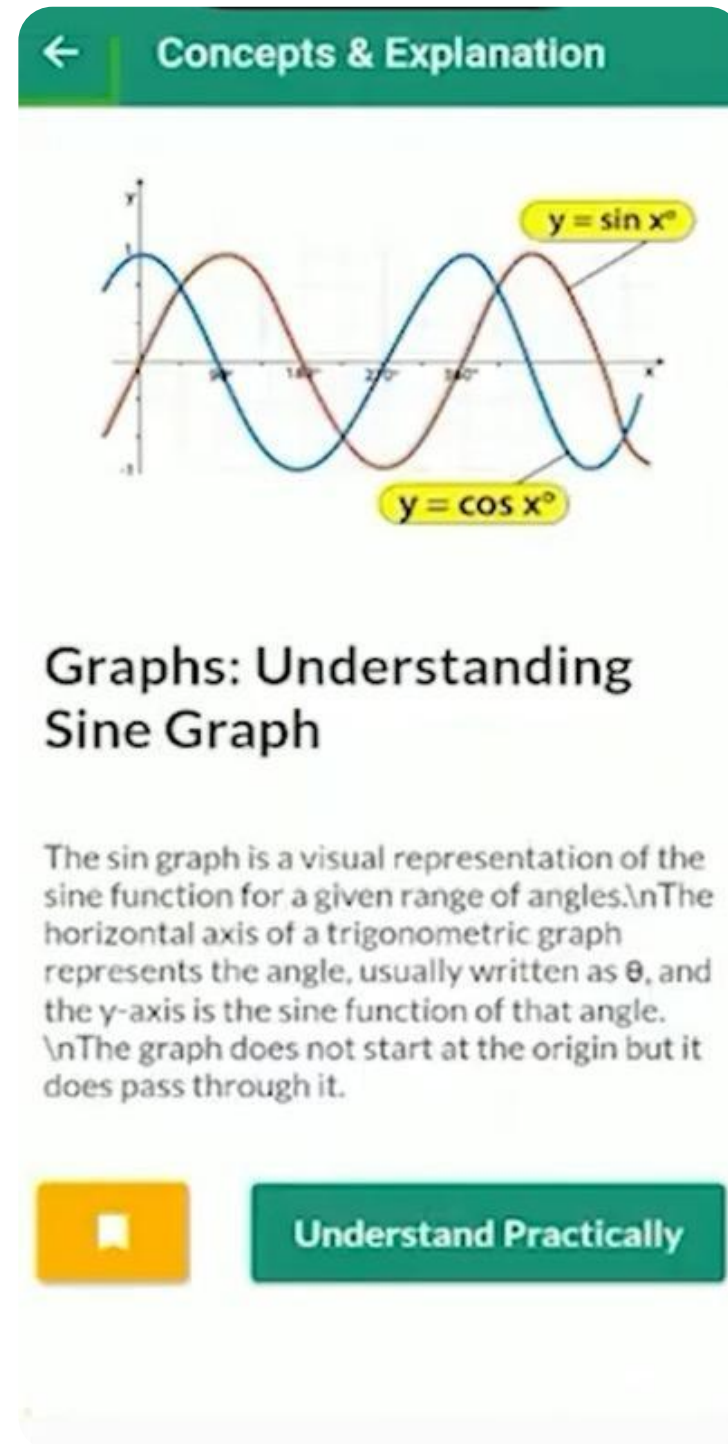


Explore and Learn various Maths and Science Concepts

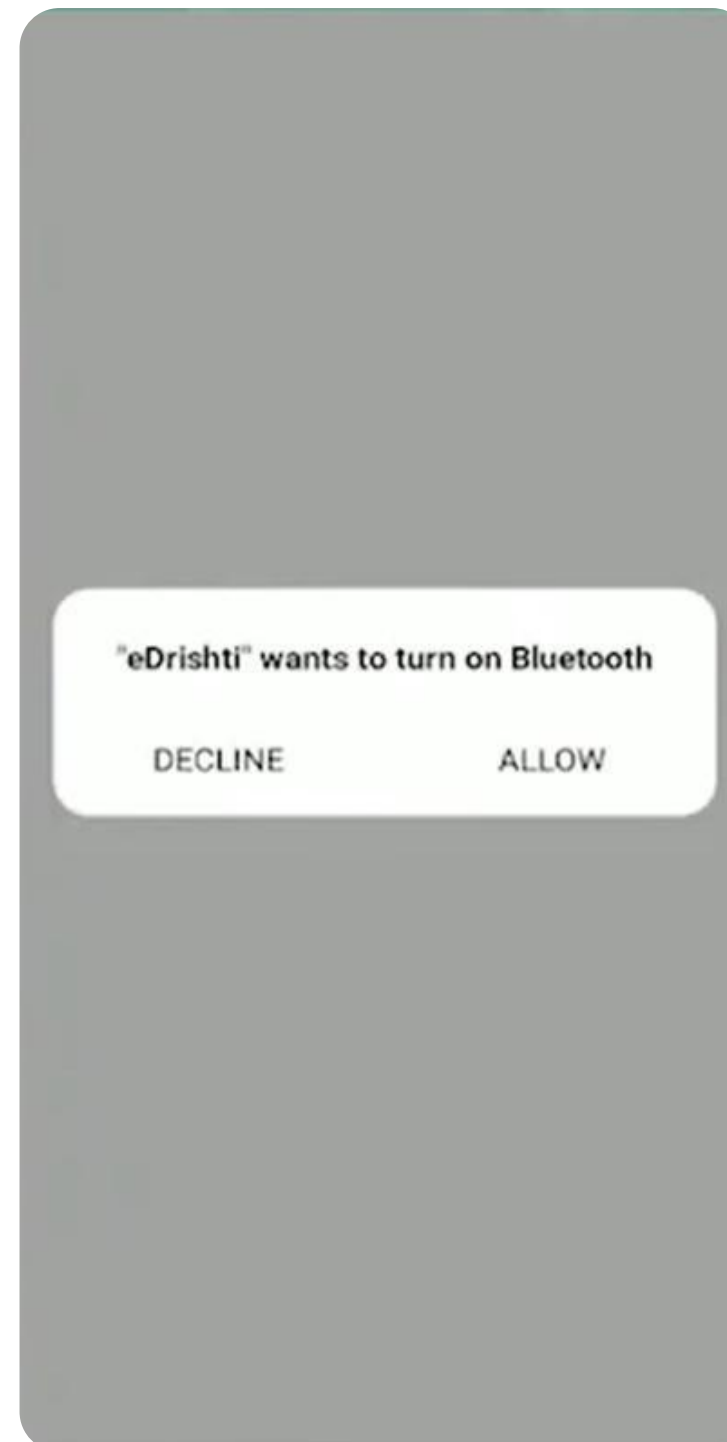




## Screens



Understanding Sine Graph



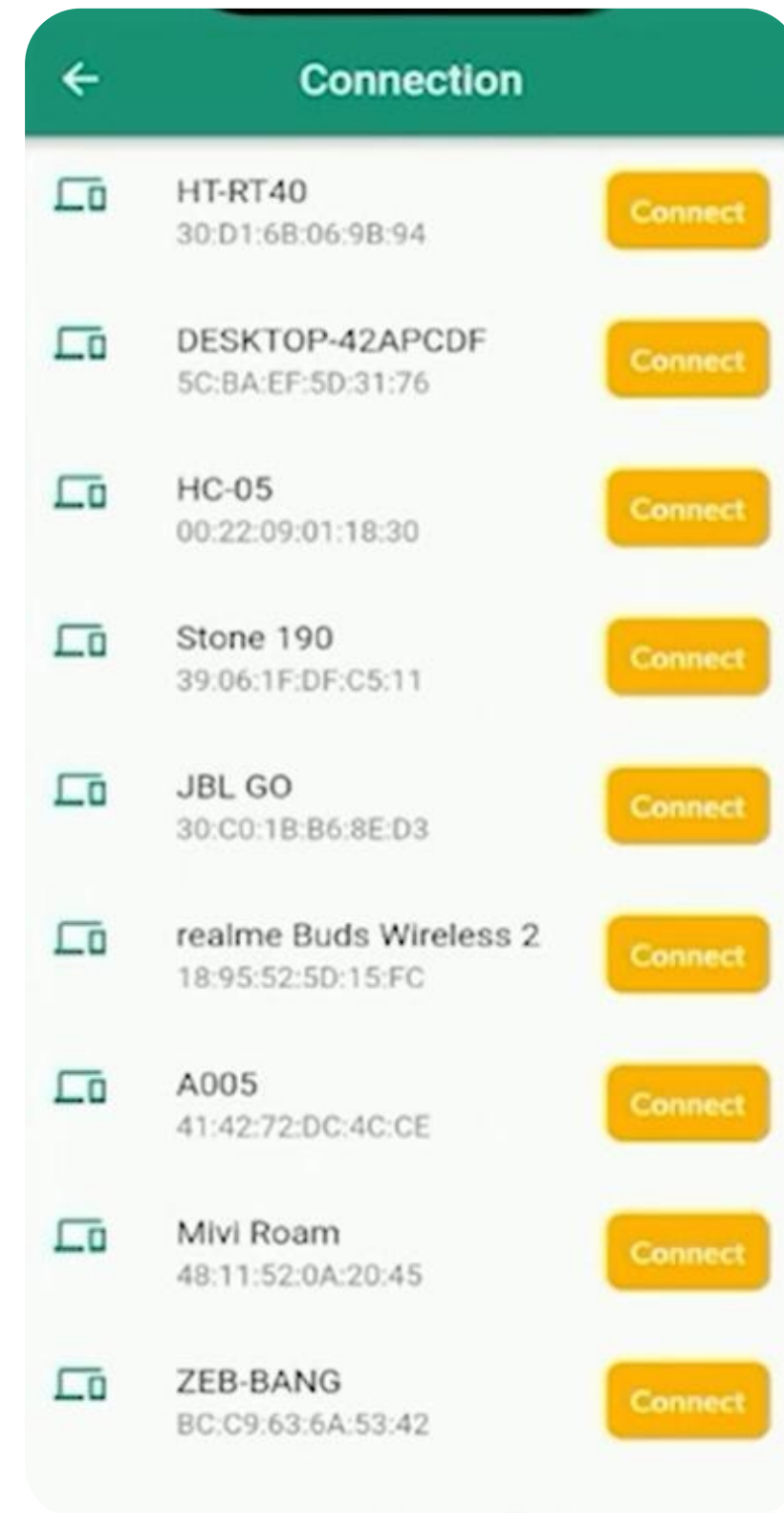
Bluetooth enable option for connecting to hardware



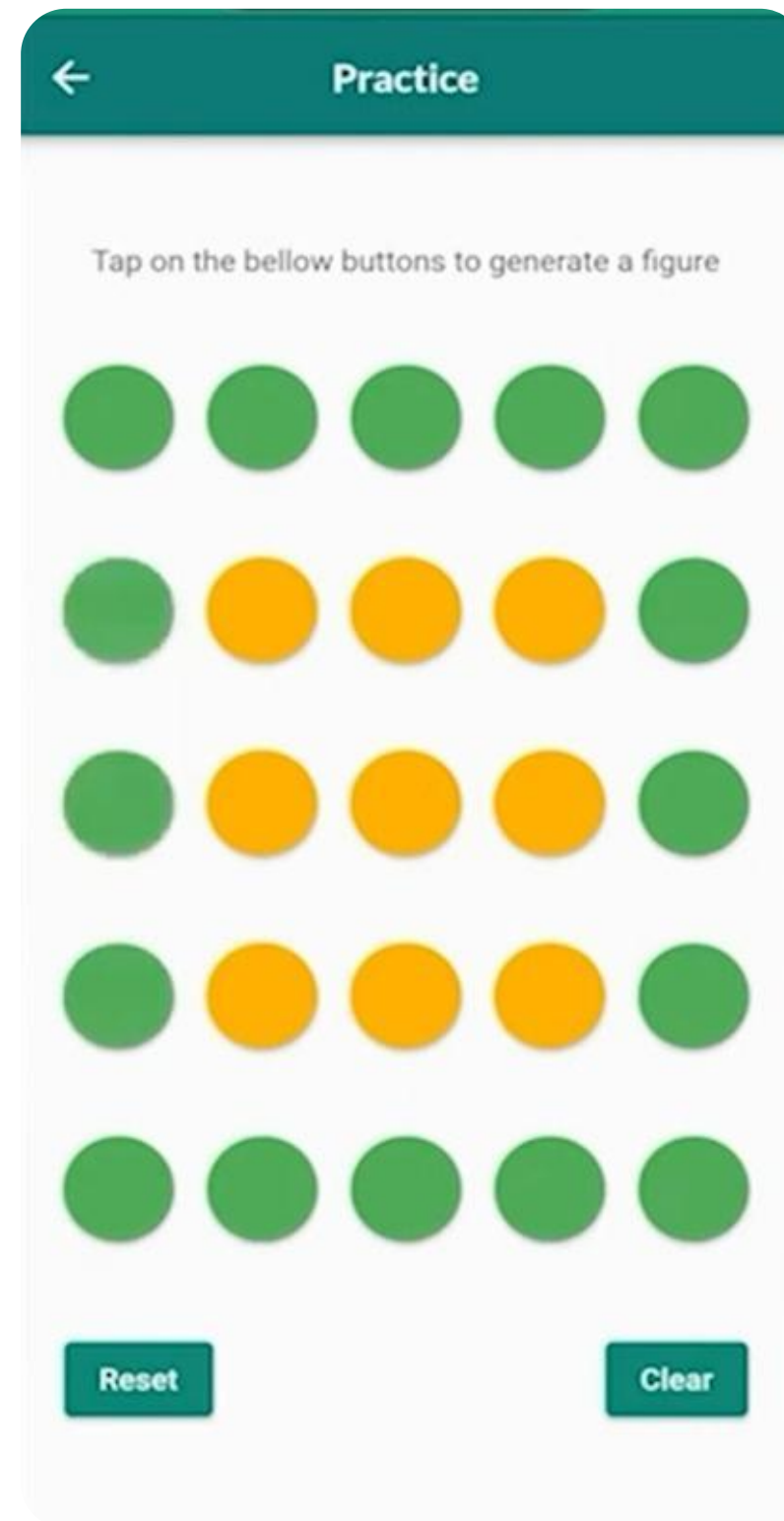
Hardware showing the practical representation of Sine Graph



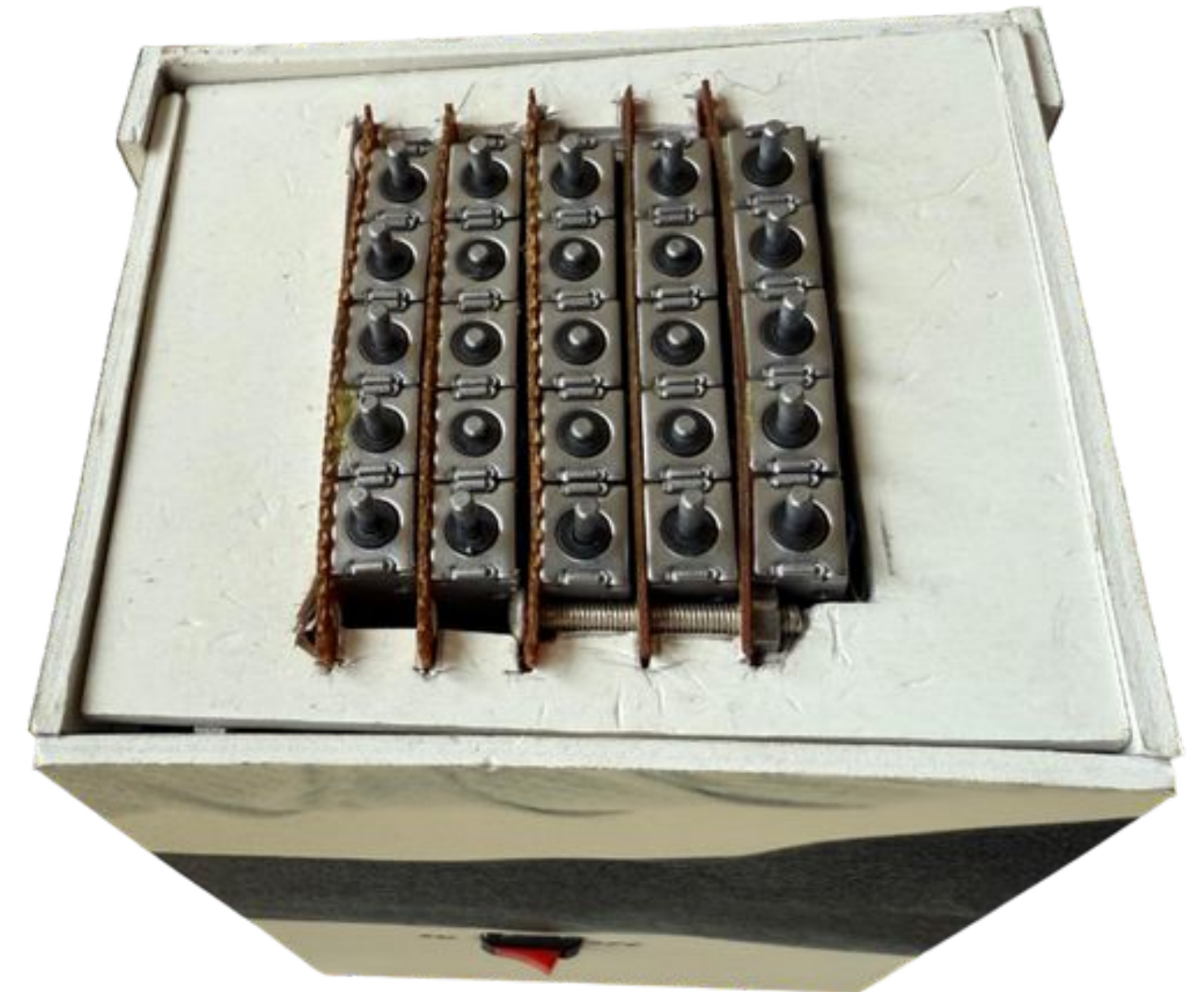
## Screens



Connecting Bluetooth



Drawing square on app

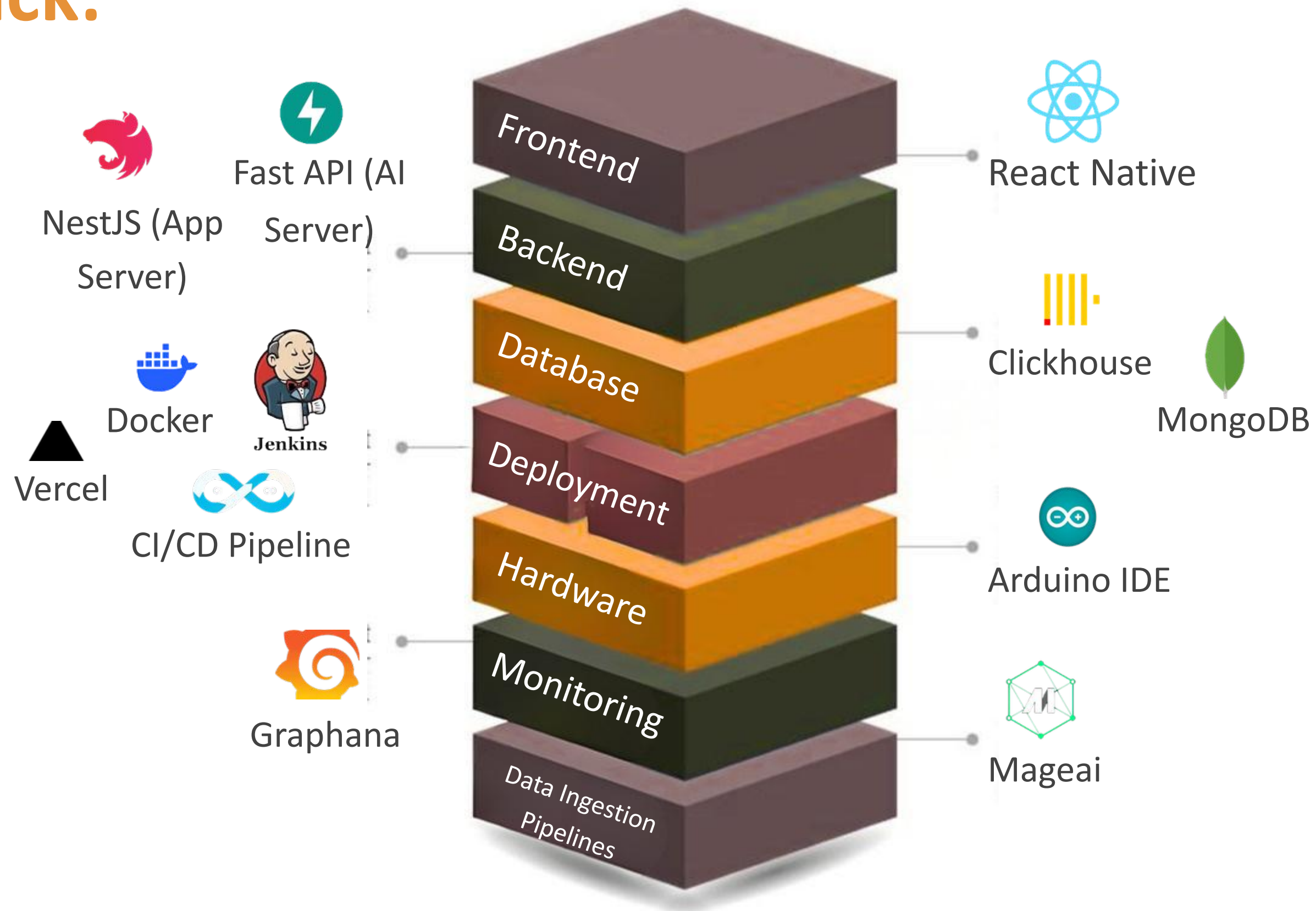


Hardware showing  
square figure



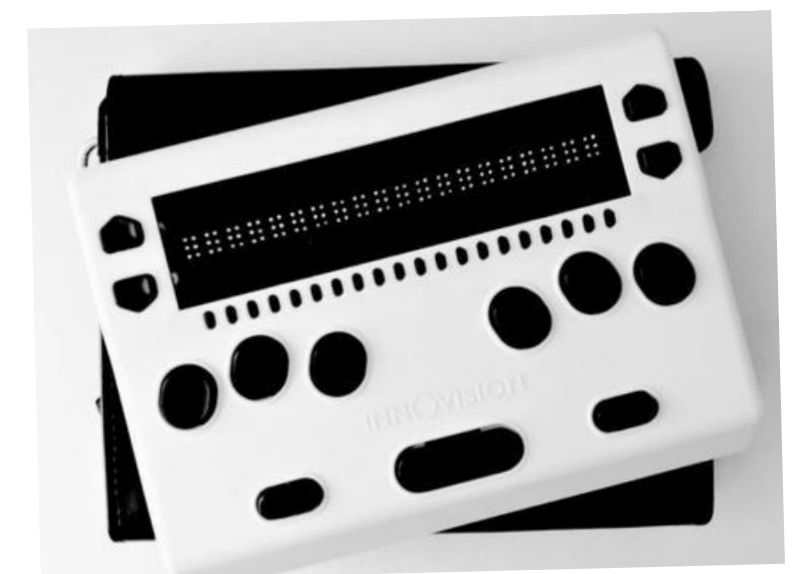


# Tech Stack:





|                                | Portability | Repairability | Cost-effective |
|--------------------------------|-------------|---------------|----------------|
| VIISAR                         | ✓           | ✓             | ✓              |
| Graphiti                       | ✓           | ✗             | ✗              |
| Dot Pad                        | ✓           | ✗             | ✗              |
| BrailleMe<br>(Innovision Tech) | ✓           | ✗             | ✓              |





**TAM**

**\$20 Billion+**

Global visually impaired  
learners who need accessible  
STEM tools

**SAM**

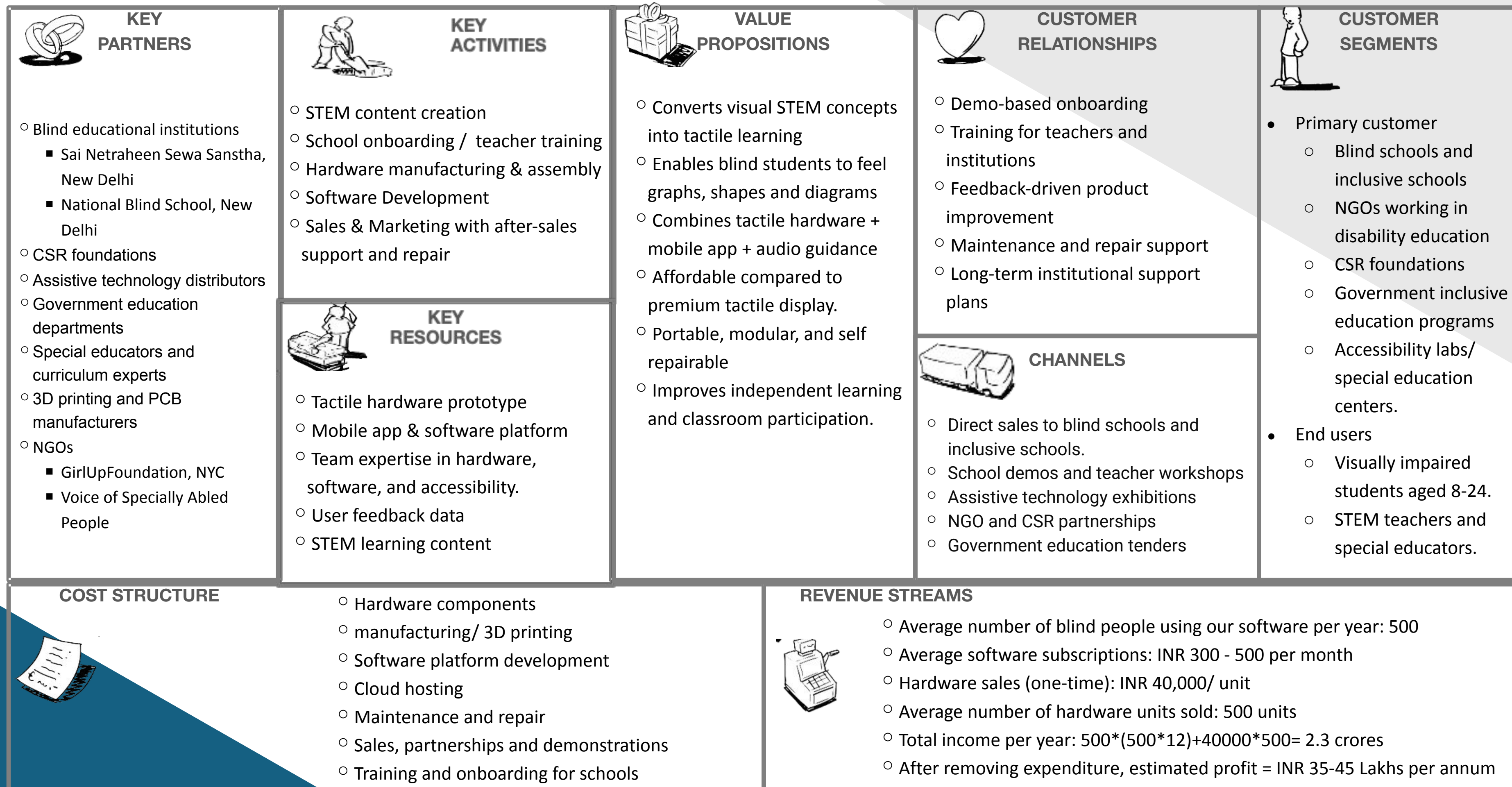
**\$200 Million+**

India + developing-market  
institutions reachable through  
schools, NGOs, CSR, and  
government

**SOM**

**\$40 Million+**

Realistic 3–5 year obtainable  
market through pilots, school  
partnerships, NGOs, and  
CSR-funded deployments.



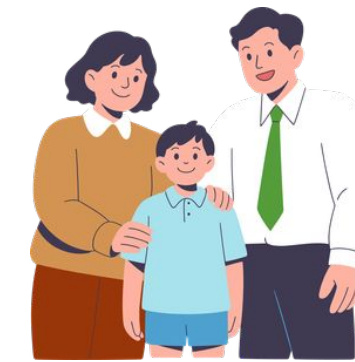




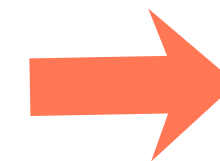
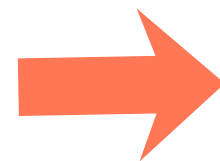
## Who is Benefited?

**Primary** - Visually impaired / blind students (especially in under-resourced schools)

**Secondary** - Teachers, special educators, schools, parents/caregivers, NGOs working in disability inclusion



## Real World Use Case:

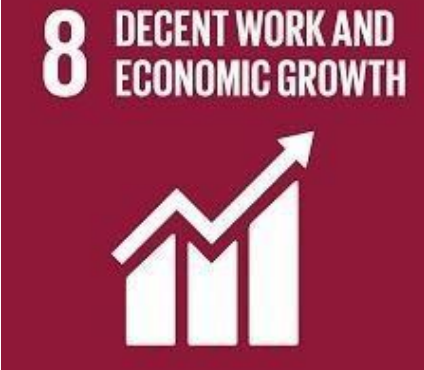


Teacher introduces topic (e.g., triangles / graphs / coordinate geometry)

Student uses the tool to feel the diagram (tactile output) + hear guidance (audio)

Student solves practice questions independently, gets feedback



| SDGs                                                                                | Direct Impact                                                                                      | Indirect Impact                                                                            |
|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
|    | Accessible STEM via tactile + audio learning; improves concept understanding and confidence        | Teacher enablement + curriculum integration for inclusive STEM classrooms and assessments. |
|   | STEM readiness → pathways to higher studies, skills development, and employability.                | Long-term economic inclusion by reducing dependency and widening career opportunities.     |
|  | Affordable assistive-tech learning infrastructure (device + app ecosystem) for learners.           | Modular design supports scalable manufacturing, local repair, and sustainable deployment.  |
|  | Cuts disability-based learning barriers; supports equal participation without constant assistance. | Scales access through schools/NGOs to underserved communities and low-income learners.     |

## Estimated Scale



**Pilot target  
(first deployment)**

3–5 schools /  
50–150  
students

**Post-pilot (after  
iteration)**

10–20 schools  
/ 300–800  
students

## Measurable outcomes



- ✔ **Learning gain:** pre/post concept test improvement (%)
- ✔ **Independence:** % tasks completed without assistance
- ✔ **Time efficiency:** time to complete diagram-based problems
- ✔ **Teacher adoption:** weekly class sessions using tool
- ✔ **Retention rate:** attendance or participation proxy
- ✔ **Cost proxy:** reduction in material dependence per learner



## Key reasons that make VIISAR the right choice

### ■ Proven Impact

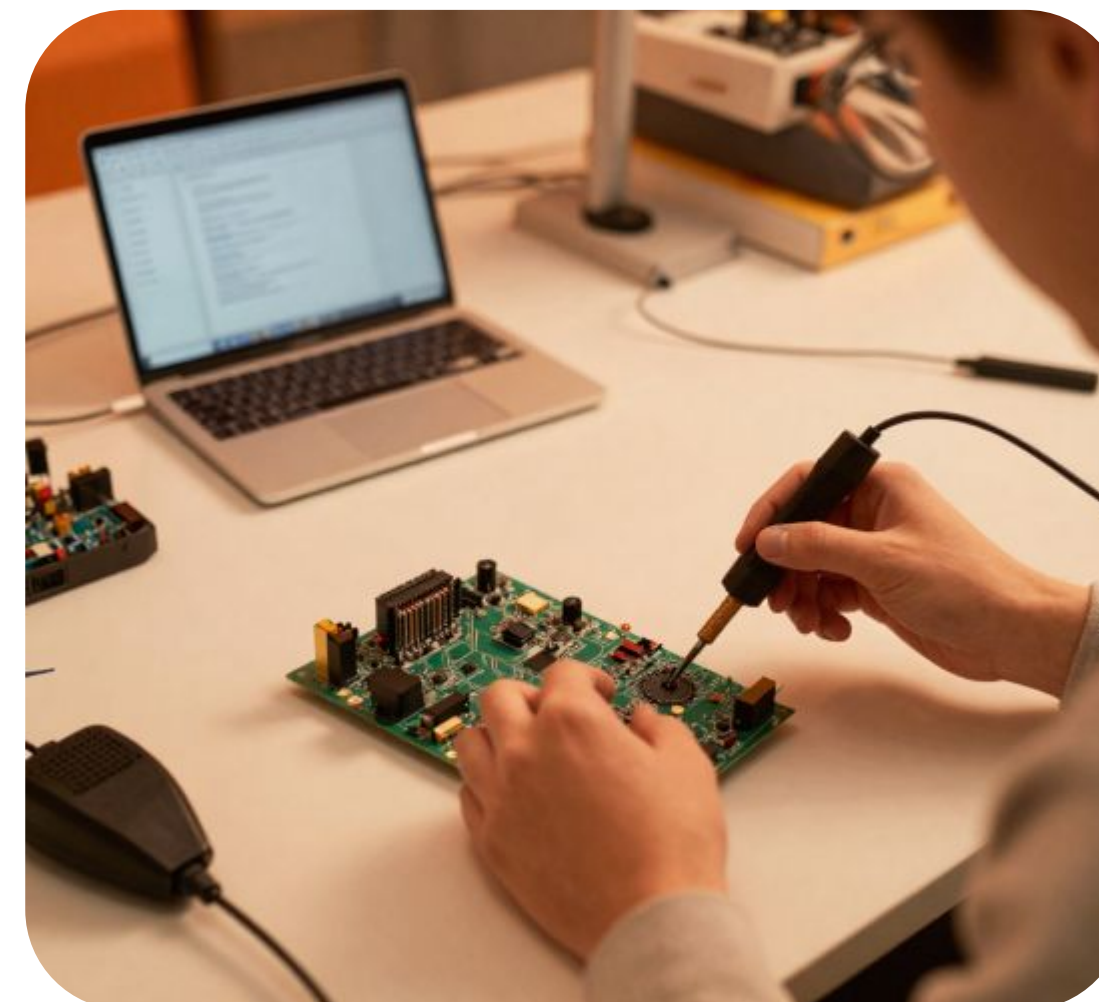
Our solution directly addresses the learning gap for 39M+ visually impaired individuals, with measurable outcomes in pilot testing.

### ■ Scalable & Affordable

Modular hardware + mobile app architecture enables rapid deployment at <\$1000/unit, making it accessible for NGOs and schools.

### ■ Strong Execution Team

Cross-functional team with deep expertise in assistive tech, embedded systems, and inclusive education design.



---

**39M+**

Lives Impacted

**95%**

Pilot Satisfaction

**<\$1000**

Per Unit Cost

**6 Month**

To Deploy



# Thank You

